

Analysis of Library Management System Transactions

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Stadtbibliothek

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Research Questions

- RQ 1: Spatial and Structural Analysis
  - How are library transactions distributed across different **locations and branches**?
  - Which libraries play a central role in the network of **inter-branch interactions**?
- RQ 2: Temporal Usage Pattern
  - How does library usage vary over **time(monthly, weekly, and daily)**?
- RQ 3: Genre and Demographic Analysis
  - How are transactions distributed across **different genres**?
  - How does genre preferences vary across **ages and genders**?

Data Overview

Time: Q4 in 2025 (2025-10-01 00:15:00 – 2025-12-31 23:40:00)

Type	owning	lending	Case	Format	Age	gender	Genre
A	Motorama	Motorama	Own	Print	19	Male	Lifestyle
R	Giesing	Giesing	Own	DVD & Bluray	57	Female	Humanities
C	Sendling	Motorama	Inter	CD	21	Male	Literature
A	Motorama	HP8	Inter	Print	42	Female	Children & Youth
T	Neuhausen	Neuhausen	Own	Print	11	Female	Children & Youth

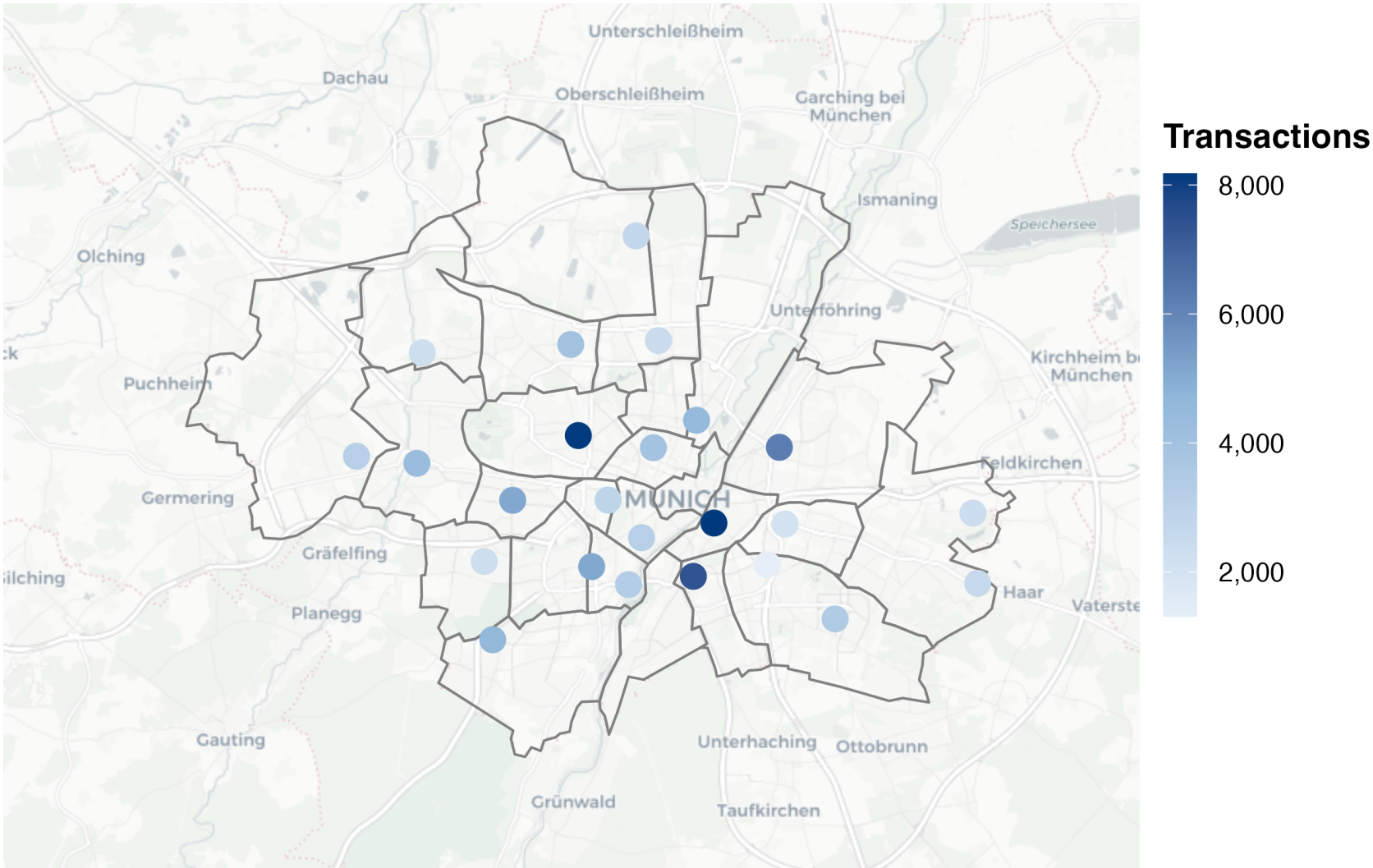
Transaction Types

A = Borrow **R** = Return
T = Renewal **C** = Rejected renewal

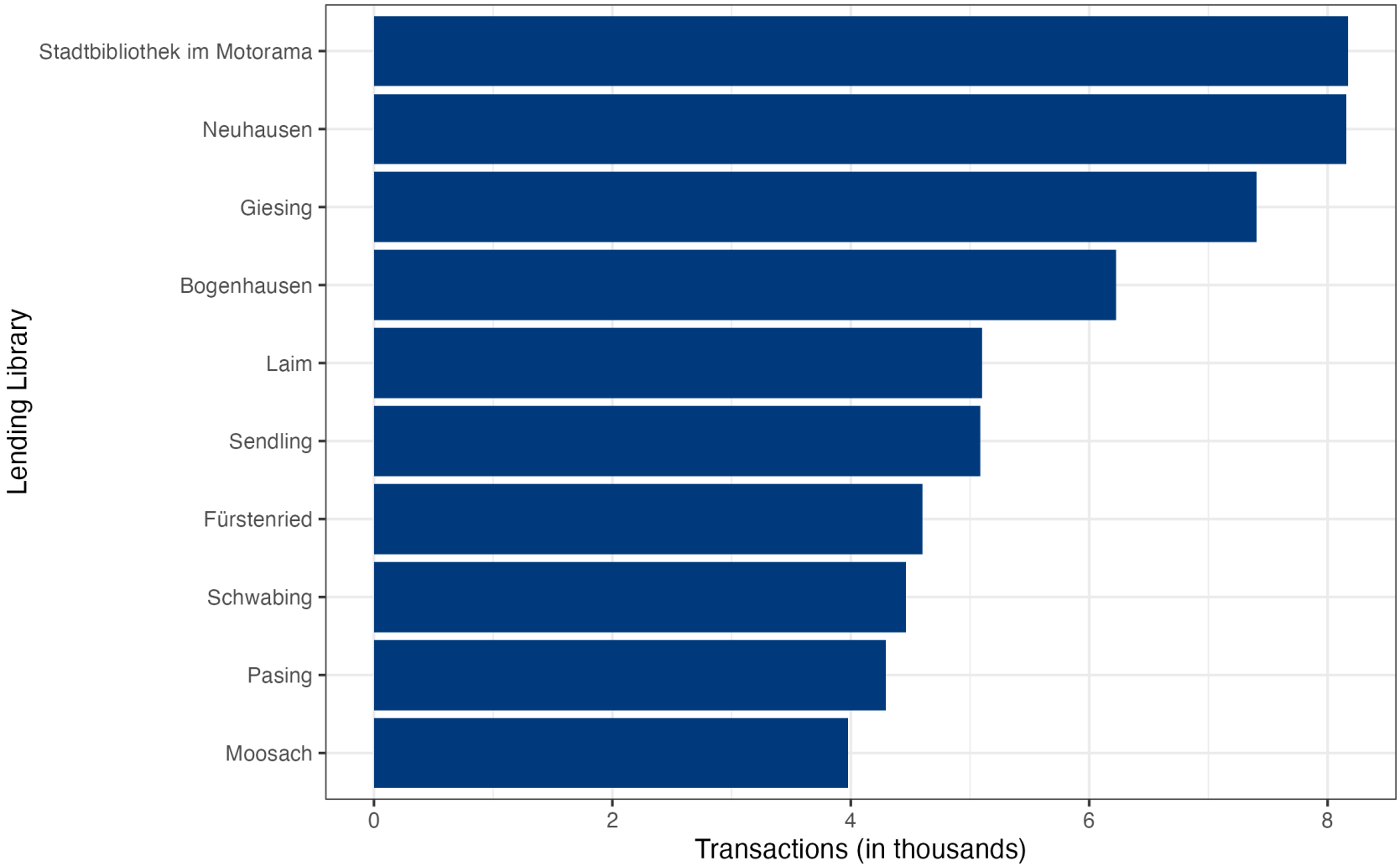
Data Size Before Cleaning: 4,689,173 rows
After Cleaning: 1,489,147 rows

Spatial and Structural Analysis

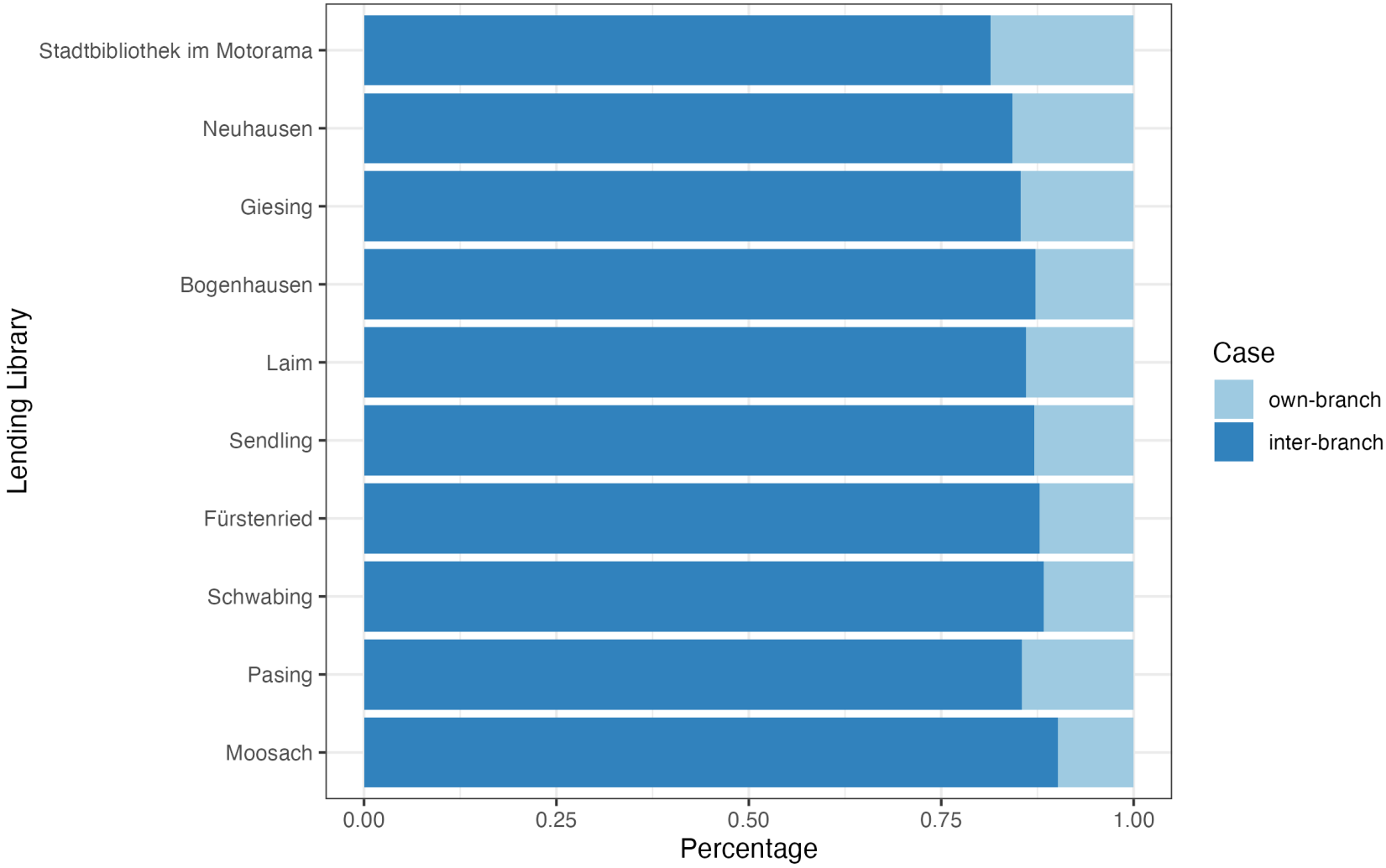
Library Usage Across Munich



Top Libraries by Number of Transactions

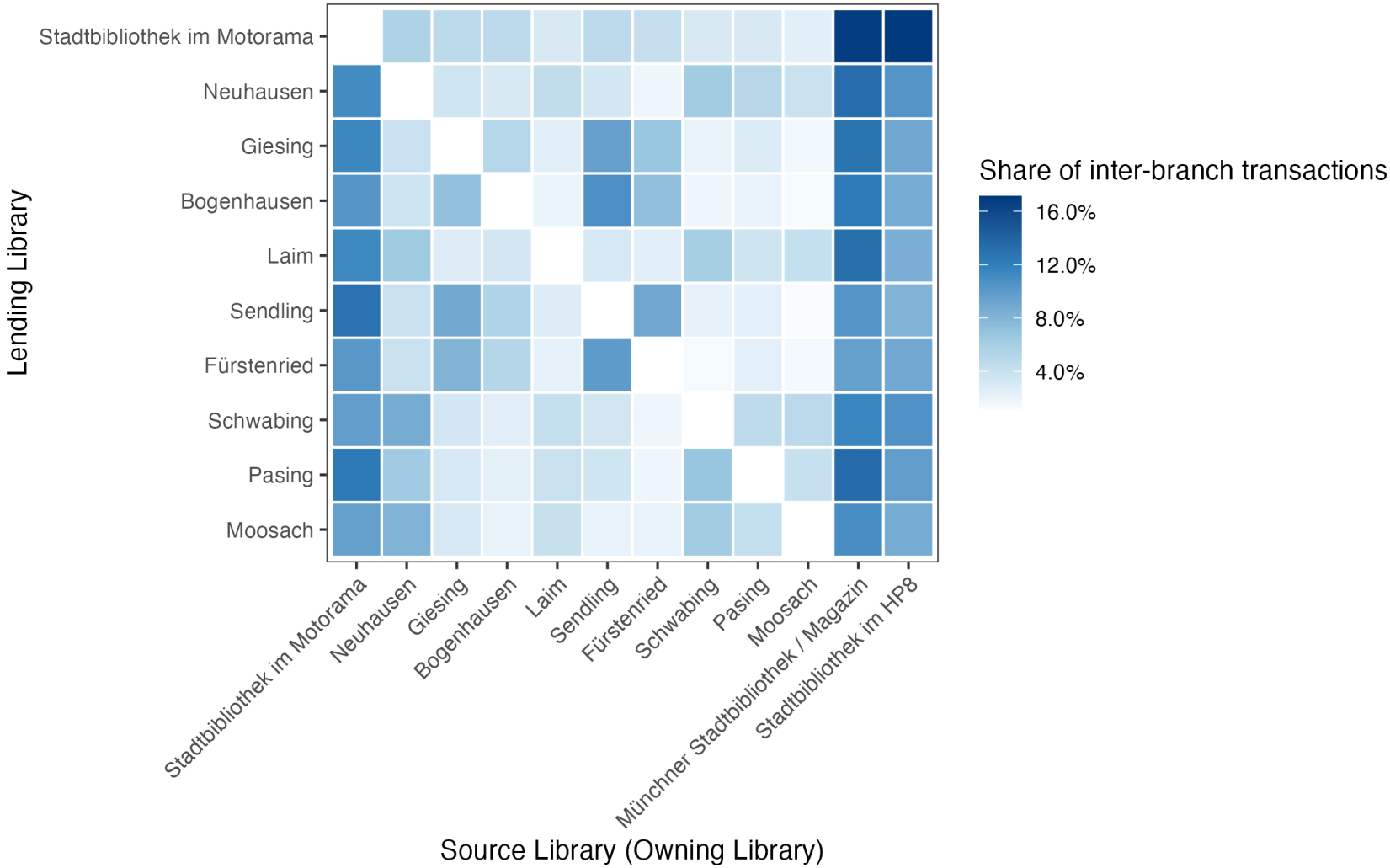


Own-Branch vs Inter-Branch Borrowing by Library



- On average, about 86% of interbranch transactions

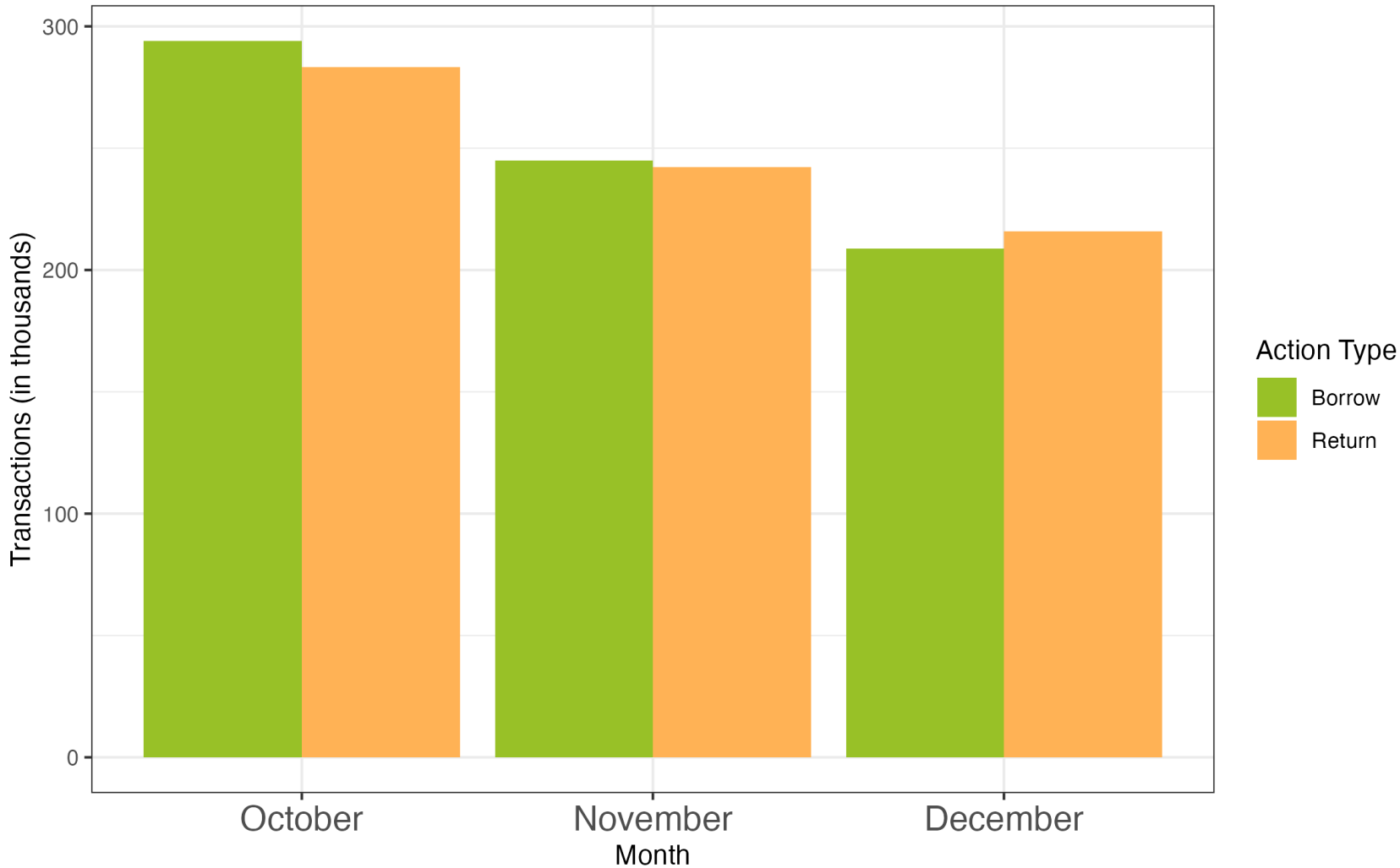
What Are the Sources of Inter-branch Transactions?



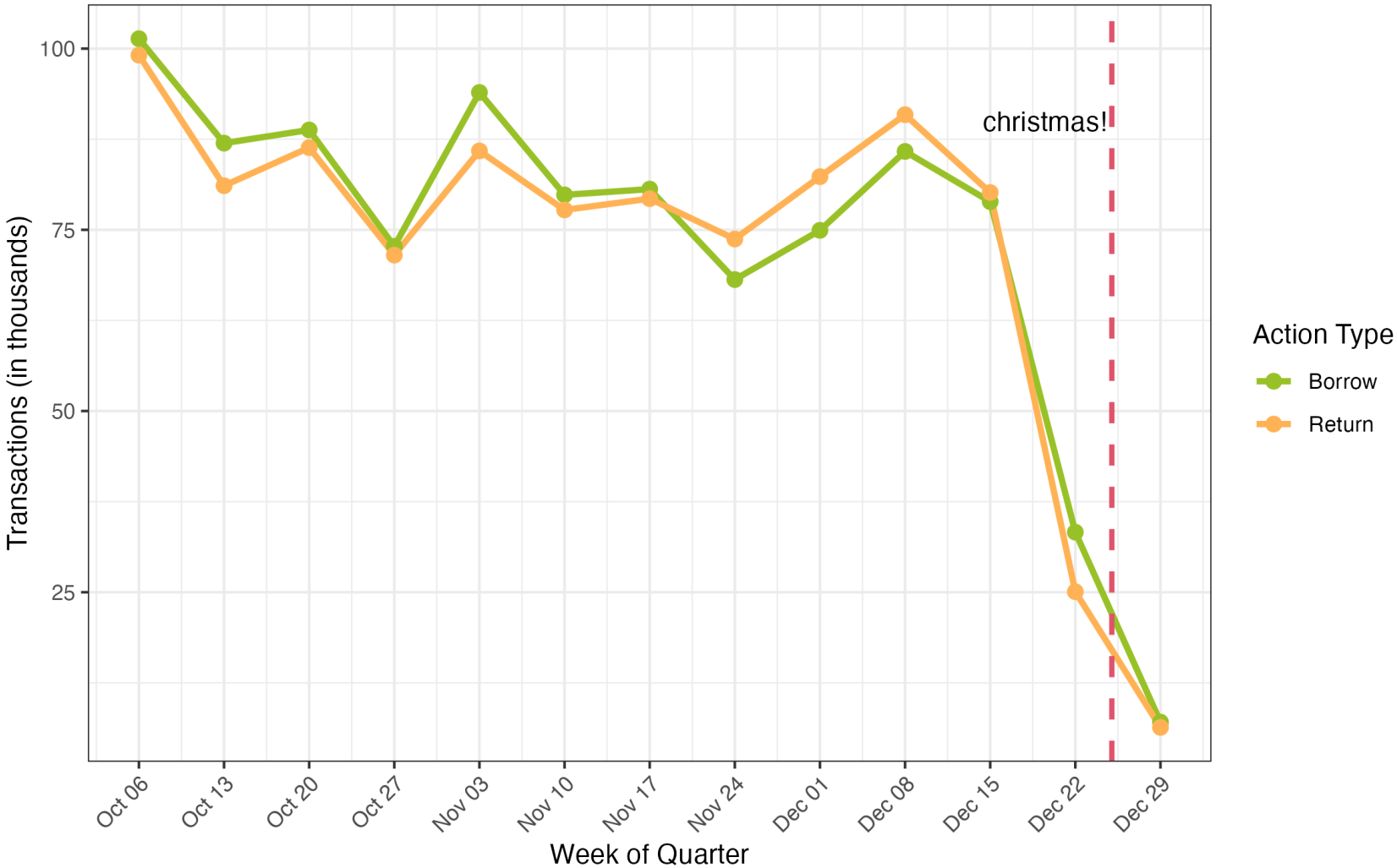
- On average, ~32% of inter-branch transactions (top 10 libraries) come from three major hubs

Temporal Analysis

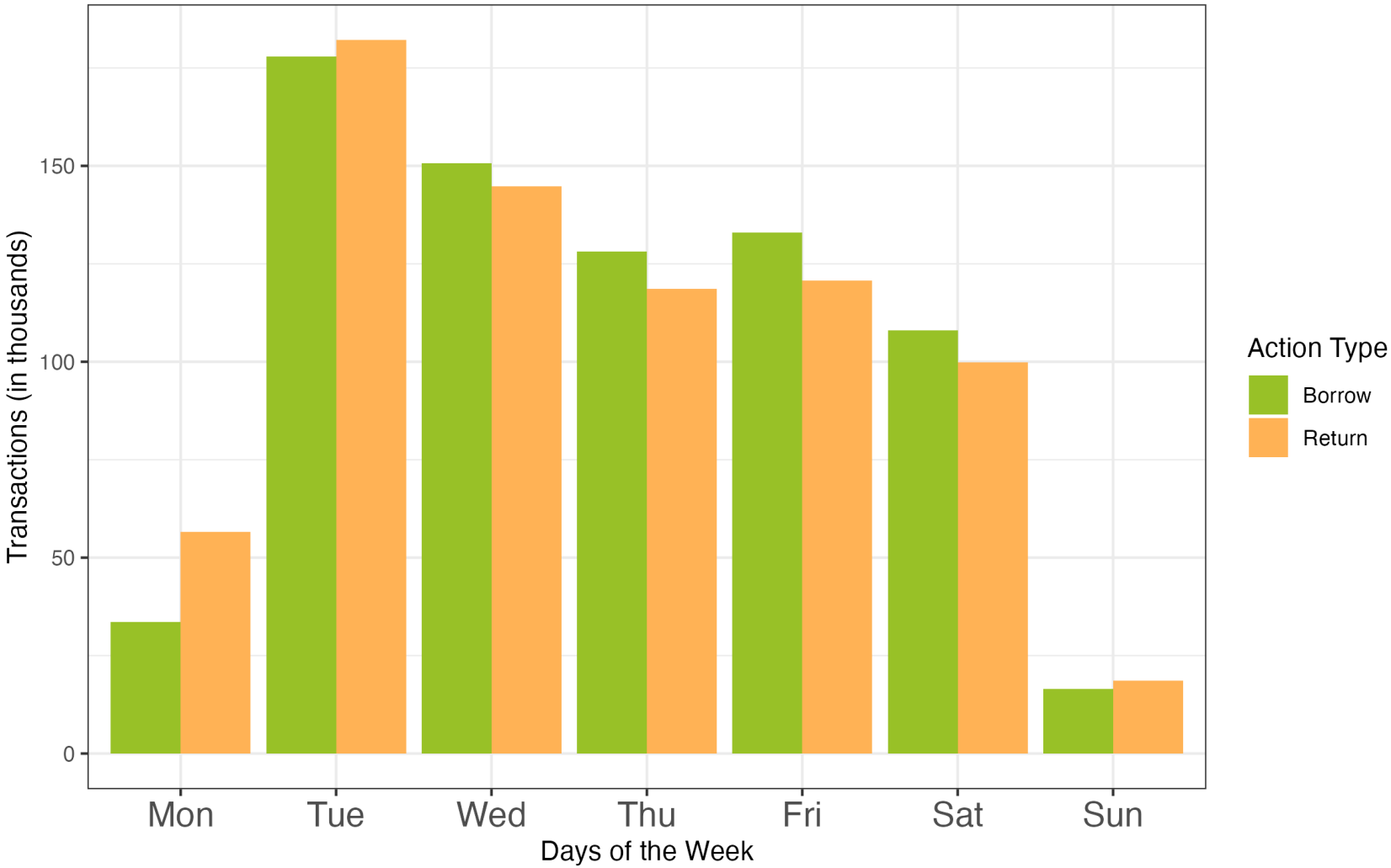
Monthly Library Transactions



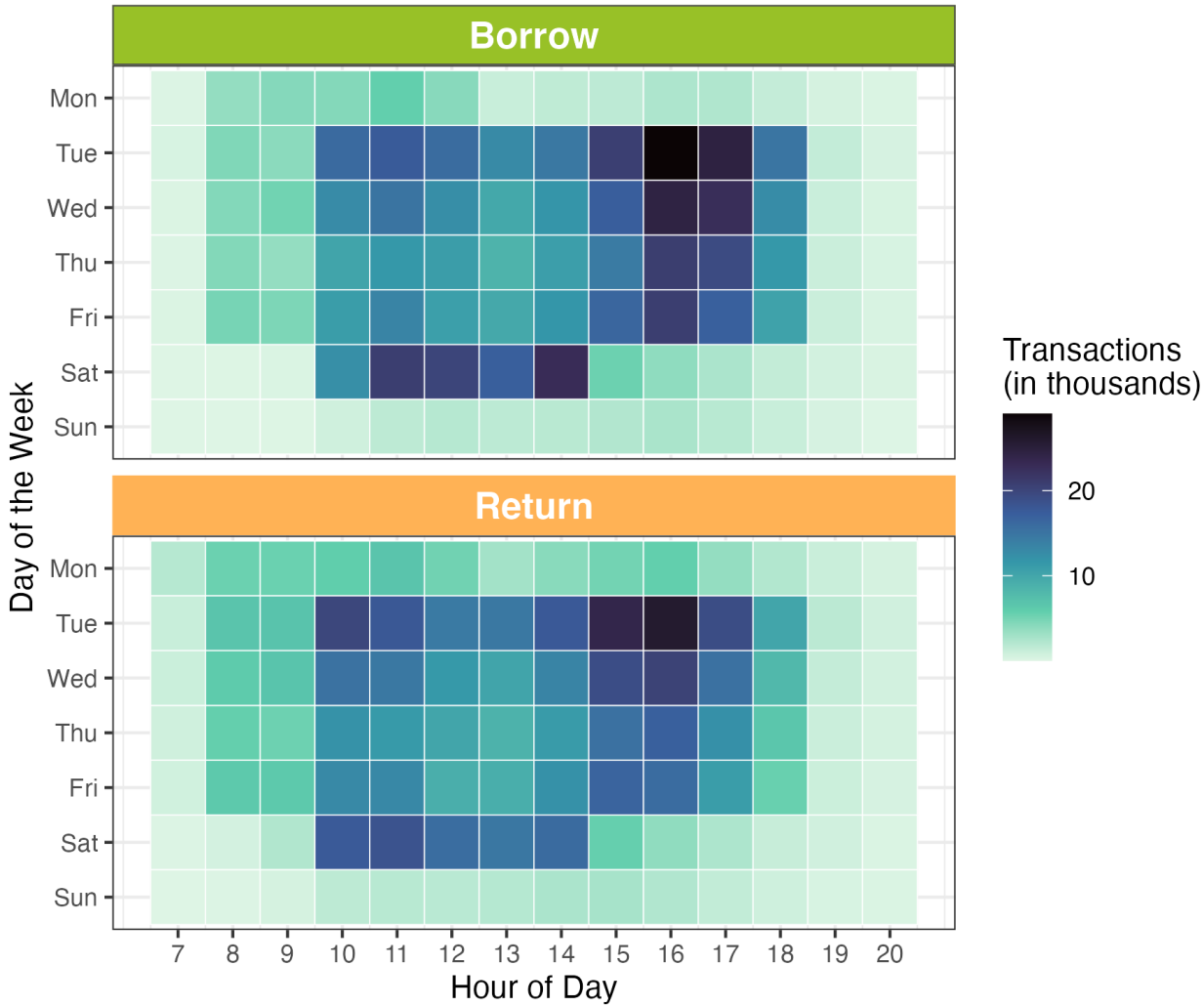
Weekly Library Transactions



Transaction Pattern by Day of the Week

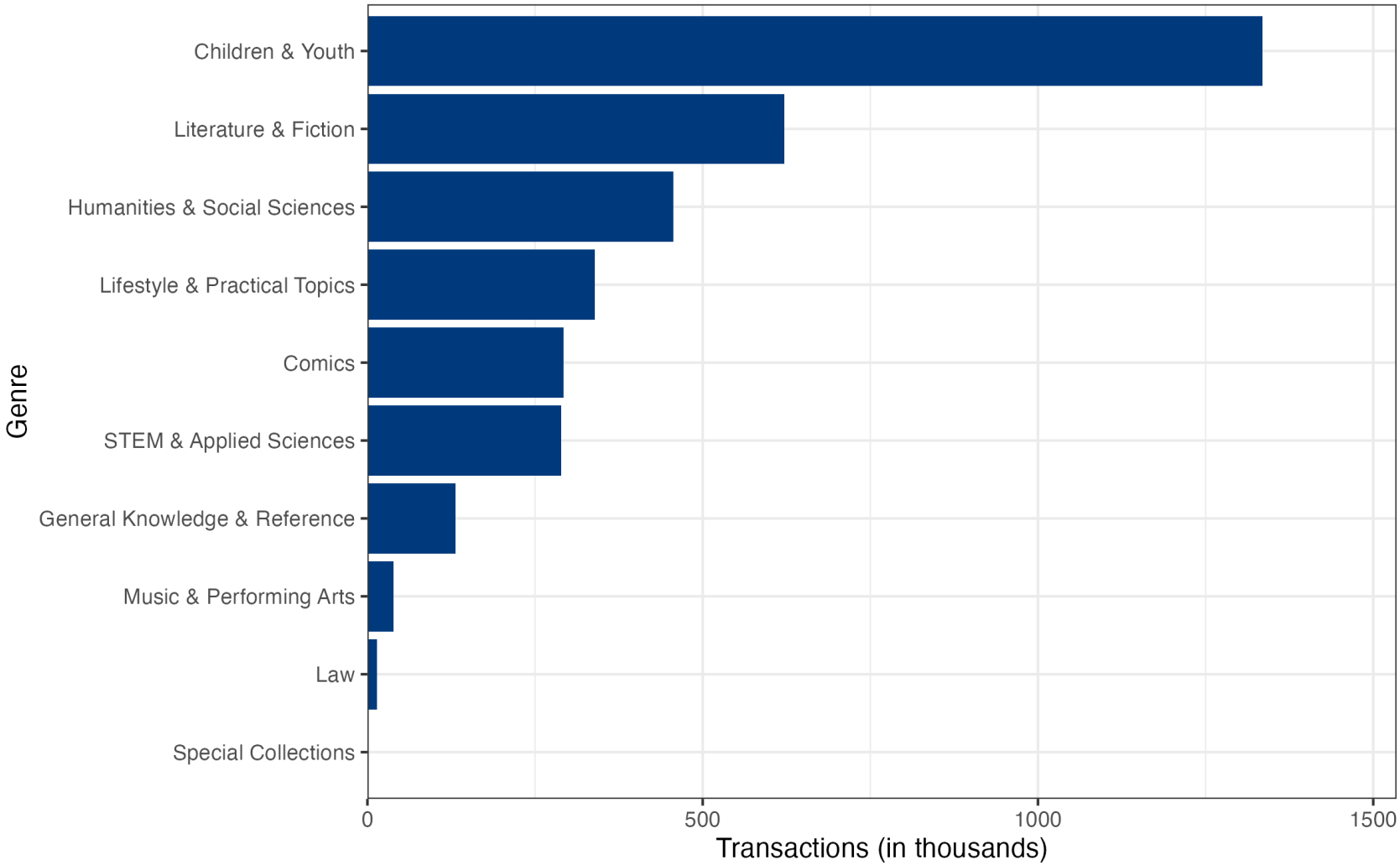


Hourly Library Transactions by Day of the Week

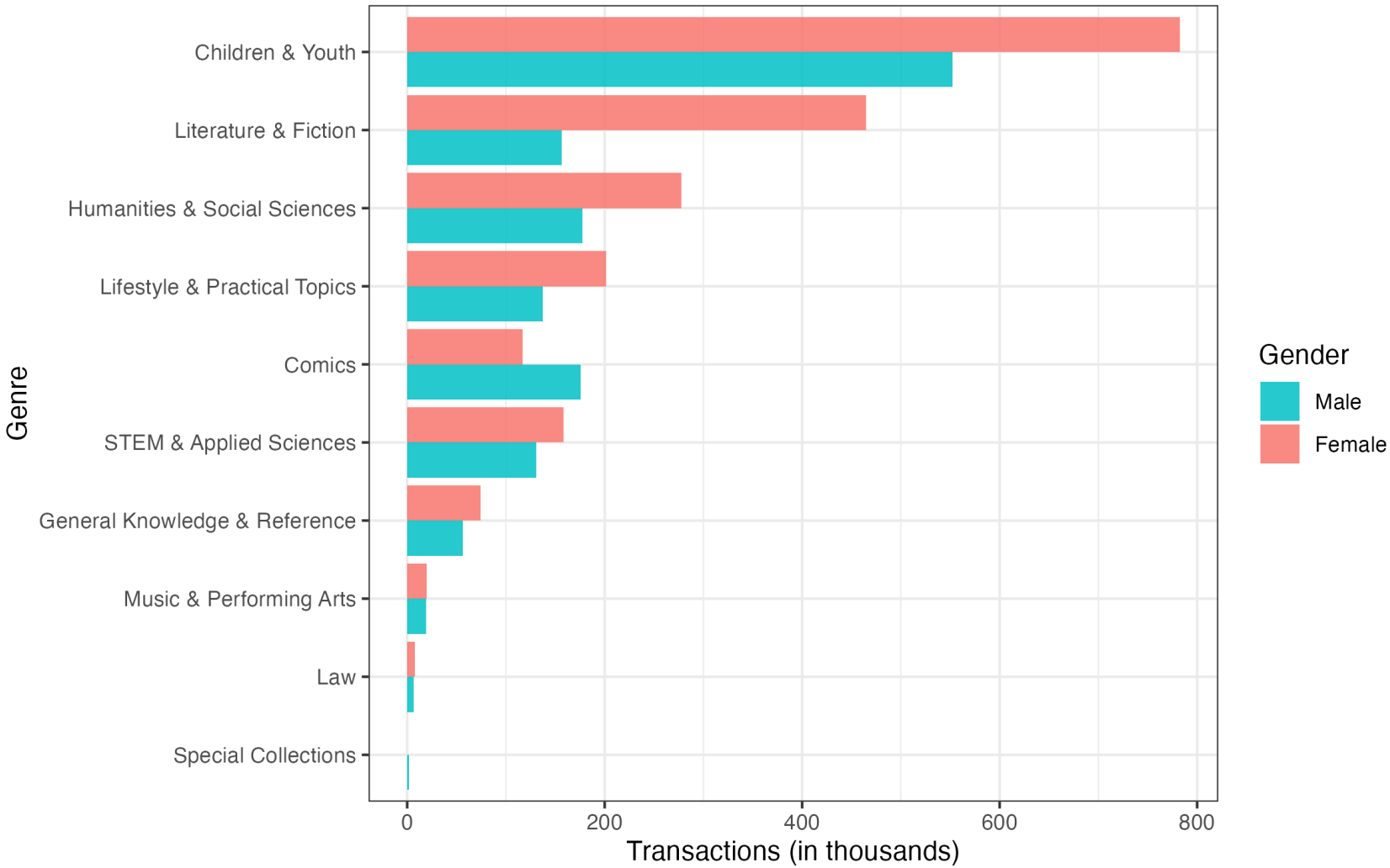


User Preference Analysis

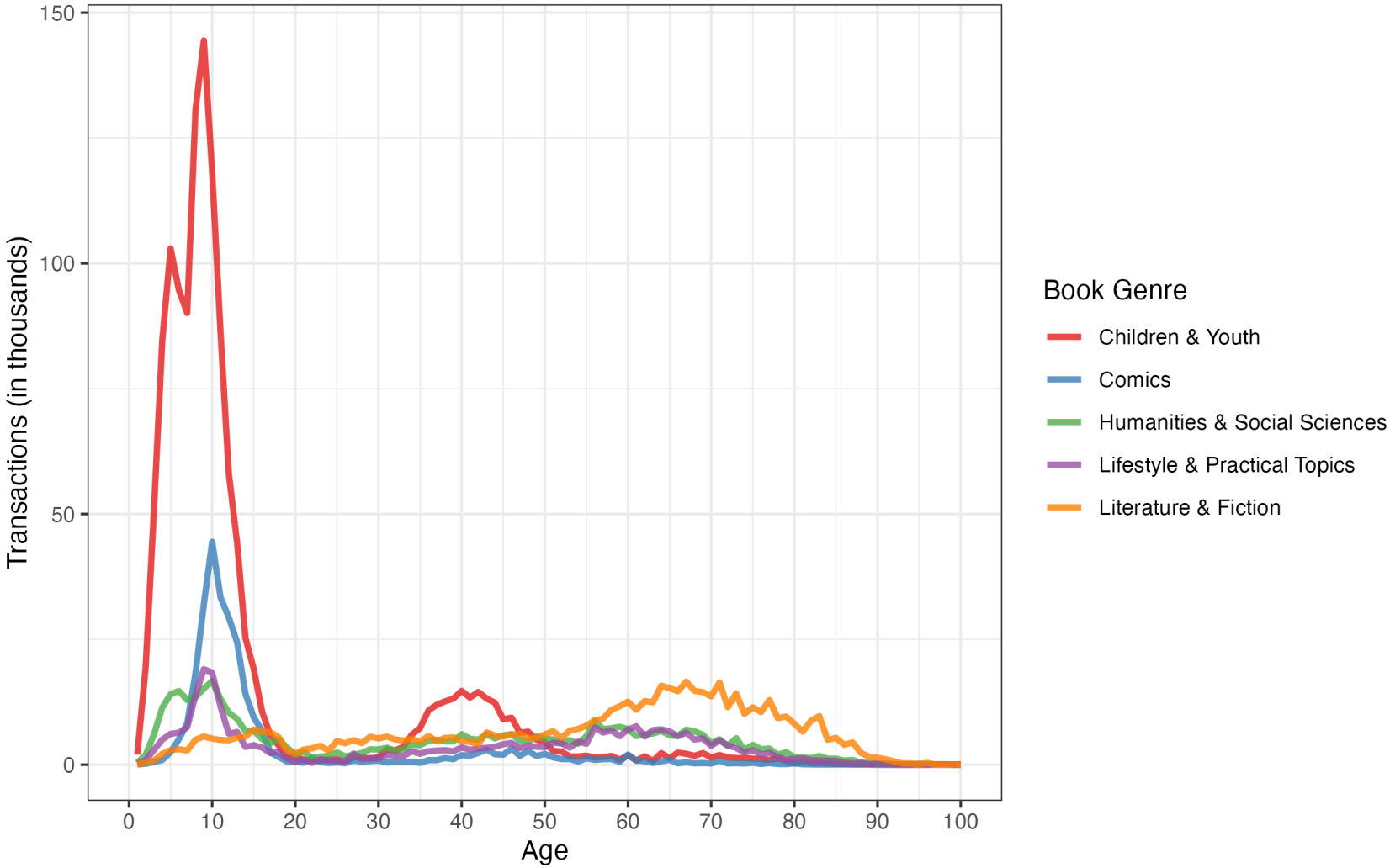
Library Borrows by Genre



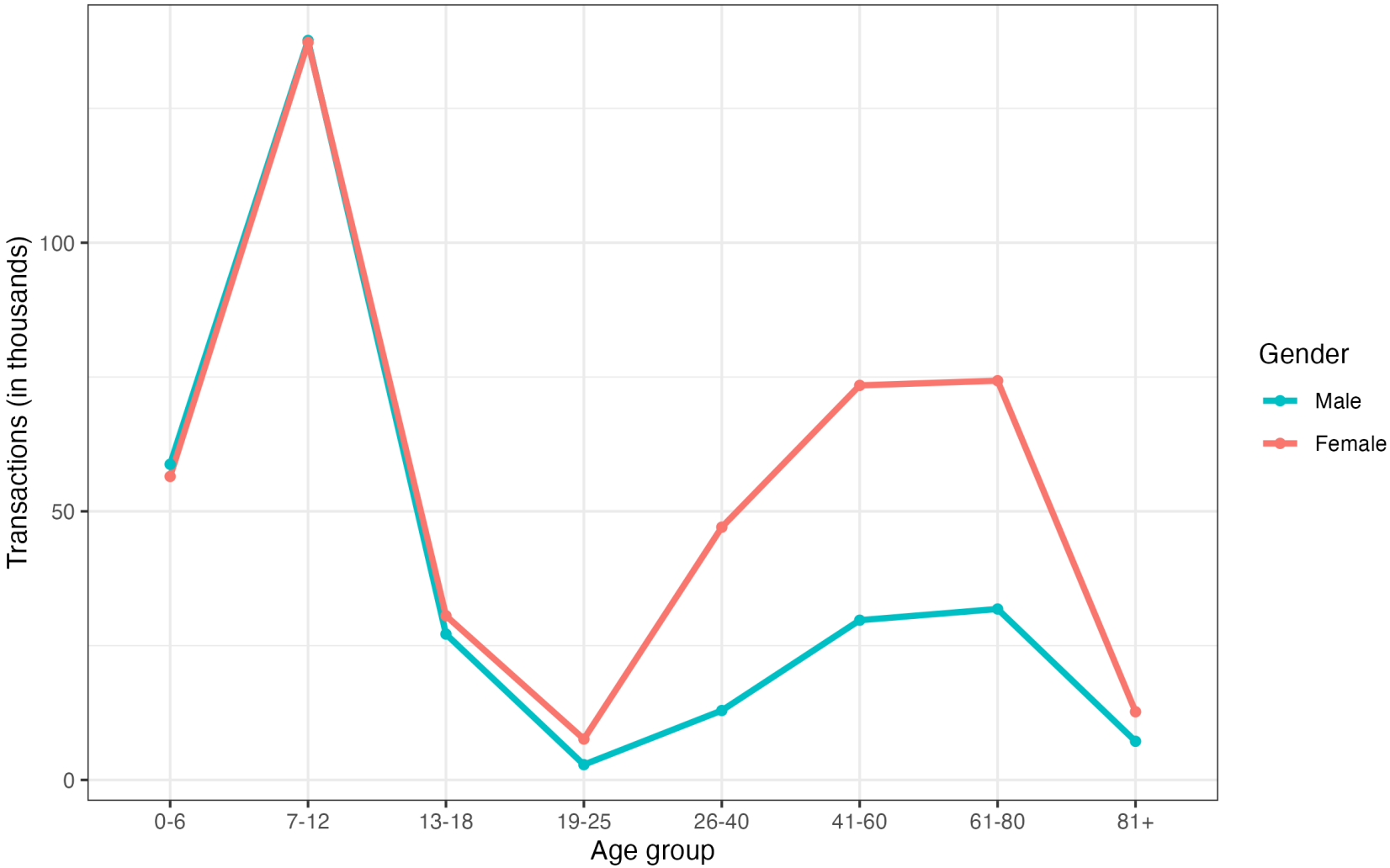
Genre Preference by Gender



Book Genre Preferences Across age Groups



Borrowing Density by Age and Gender



Key Findings and Conclusion

- Libraries located in central areas are used more frequently.
- The Münchner Stadtbibliothek, HP8, and Motorama act as major hubs for interbranch transactions, but borrowing sources remain diverse.
- A noticeable decline in activity occurs during the Christmas season (After 15th of Dec).
- Tuesday showed the highest level of usage, especially between 3pm and 5pm.
- Users aged 7 to 12 represent the most active group (Children & Youth).
- Genre preferences vary across different age groups and genders.

Limitation

- The lending library ID contains many invalid values (-1).
  After removing these entries, only about 14% of the original data remained (~86% excluded).
  This mainly affects the spatial analysis (Part 1), while other analyses are not impacted.
- The dataset only covers Q4 2025 (Oct –Dec).
  Seasonal patterns across other periods cannot be analyzed.
- Detailed genres were grouped into 10 broader categories.
  This may reduce granularity.
- Categories such as diverse and institute (~11%) were excluded.
- The analysis focuses only on print materials, excluding other media types (~ 25%).

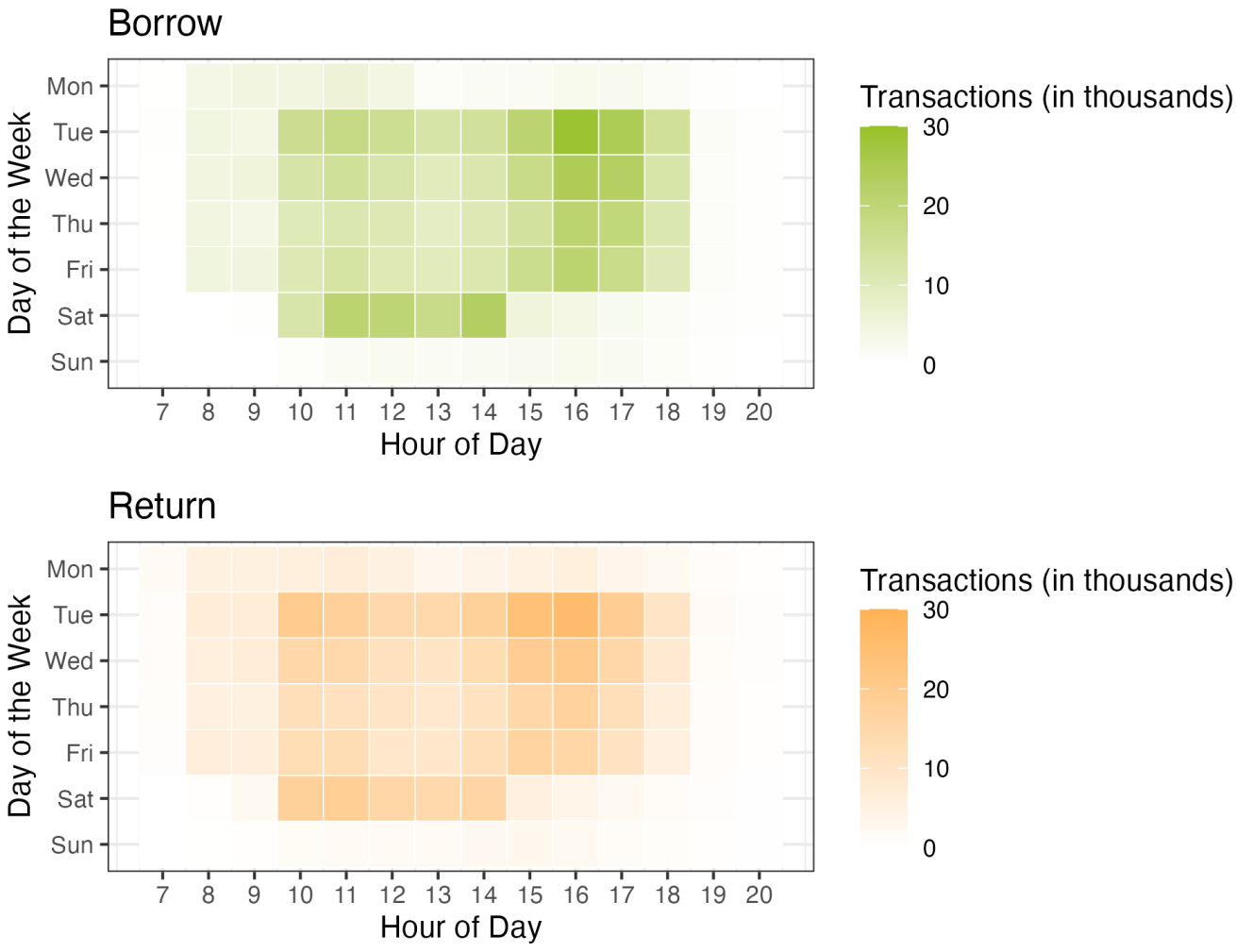
Discussion

- A small number of central libraries (e.g., Motorama, HP8, Magazin) act as key hubs, but inter-branch borrowing remains widely distributed across the network
- Library usage follows clear temporal patterns, with strong weekday peaks (especially Tuesday afternoons) and a sharp decline during Christmas
- User behavior is highly demographic-driven: younger users (7–12) dominate activity, and genre preferences vary by age and gender

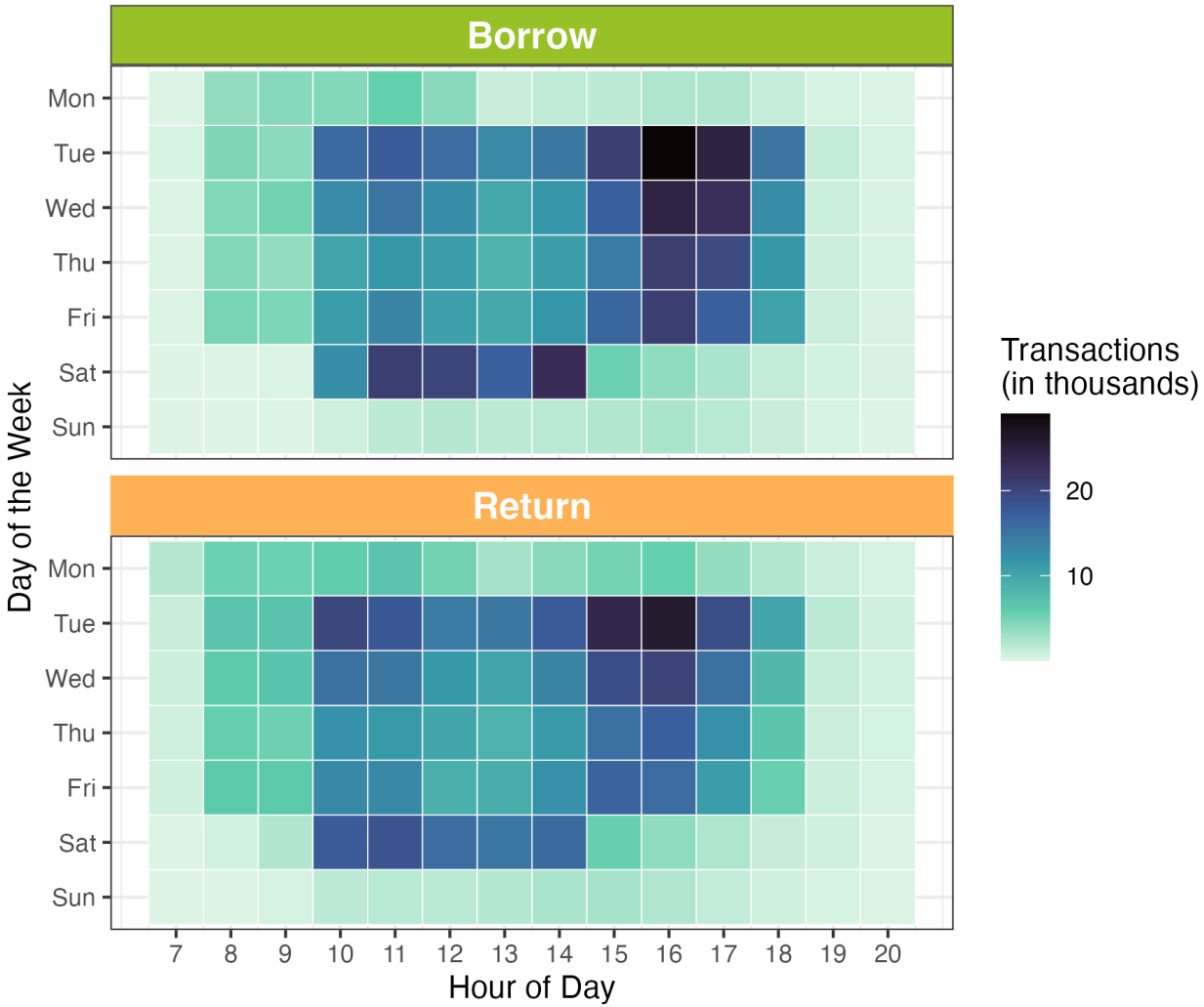
$\rightarrow$ How can these patterns be used to optimize resource allocation, staffing, and collection distribution across libraries?

Appendix

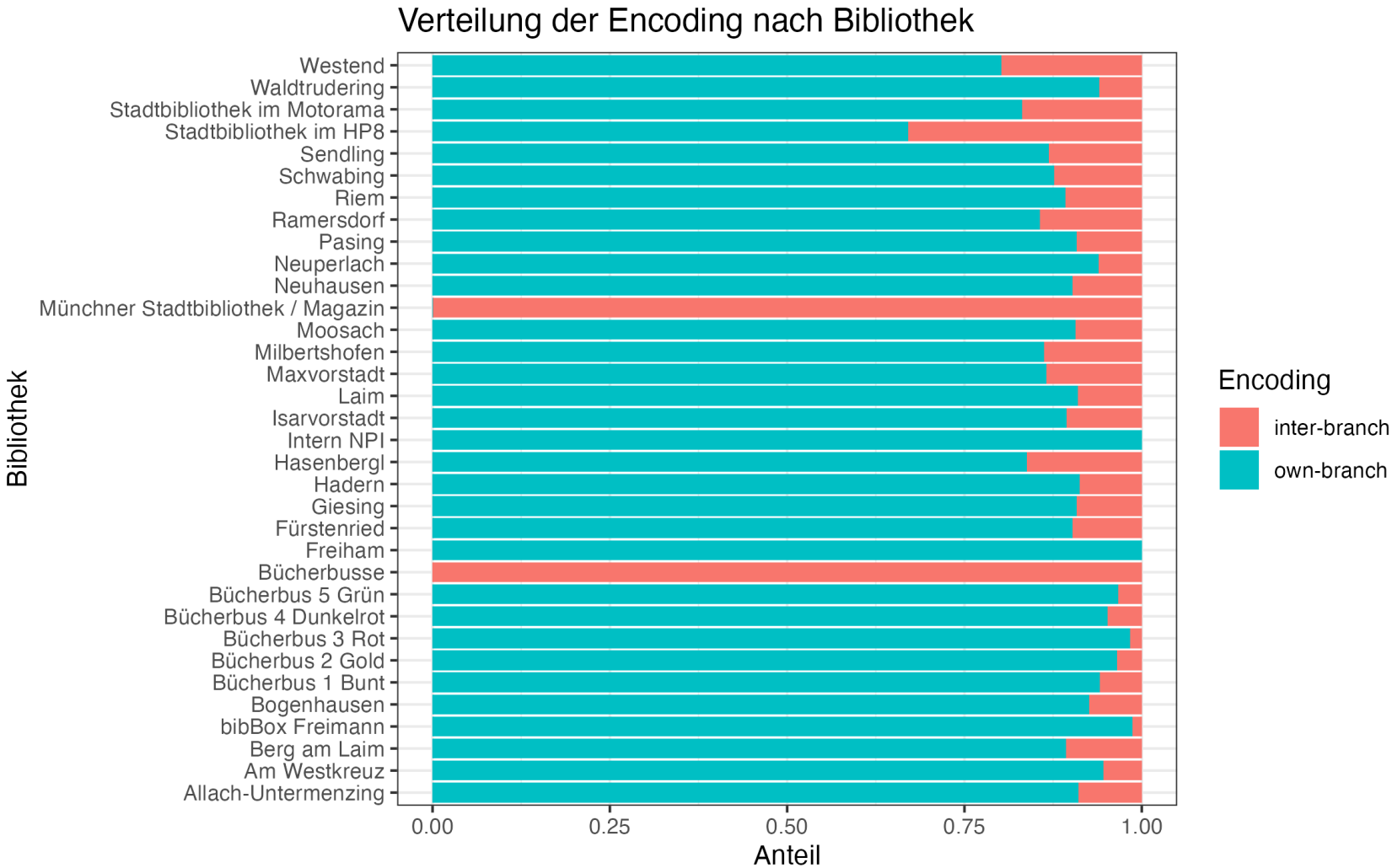
Alternatives : 2_4



Alternatives: 2_4



Alternatives: Interbranch vs Ownbranch (Proportion)



Alternatives: Full Heatmap between libraries

